

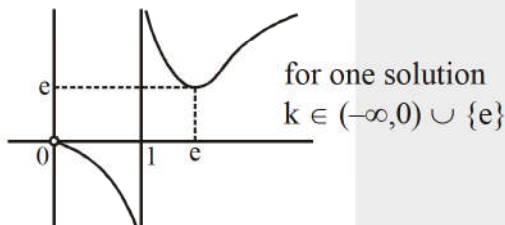


WORK SHEET - 19

APPLICATION OF DERIVATIVE

Q.1 (C)

$$f(x) = \frac{x}{\ln x} = k$$



Q.2 (A)

$$\frac{dy}{dx} = -\frac{u^2}{x^2} + \frac{v^2}{(u-x)^2} = 0 \Rightarrow x = \frac{u^2}{u \pm v}$$

$$\text{So only } x = \frac{u^2}{u+v} \in (0, u)$$

$$\text{and at this value } \frac{d^2y}{dx^2} > 0 \Rightarrow \text{minima occur at } x = \frac{u^2}{u+v}$$

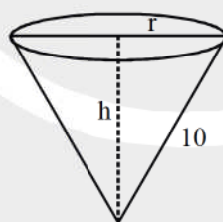
$$\text{and minimum value} = \frac{1}{u}(u+v)^2$$

Q.3 (C)

$$v = \frac{\pi}{3} r^2 h = \frac{\pi}{3} (100 - h^2) \cdot h$$

$$\frac{dv}{dh} = 0 \Rightarrow 100 = 3h^2$$

$$h = \frac{10}{\sqrt{3}}$$



Q.4 (C)

$$y' = \frac{\ln x - 1}{(\ln x)^2}$$

Since $(\log x)^2$ is always positive, we have $y' > 0$ if $x > e$. Therefore y increases on $[e, e^3]$.
Therefore greatest value occur at $x = e^3$.

$$\text{Hence } u = \frac{e^3}{3} \Rightarrow \frac{e^3}{u} = 3. \text{ Ans.}$$



Q.5 (B)

$$y = |x| + a$$

$$y = |\sin 2x|$$

$$\frac{dy}{dx} = 2 \cos 2x = 1$$

$$\cos 2x = \frac{1}{2}$$

$$2x = \frac{\pi}{3} \Rightarrow x = \frac{\pi}{6}$$

$$\Rightarrow p\left(\frac{\pi}{6}, \sin \frac{\pi}{3}\right)$$

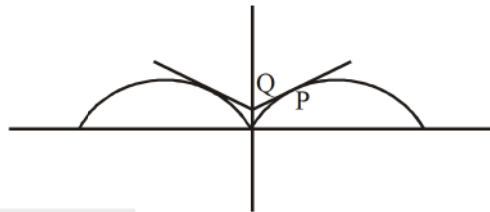
Equation PQ :

$$y - \frac{\sqrt{3}}{2} = \left(x - \frac{\pi}{6}\right)$$

$$y = x - \frac{\pi}{6} + \frac{\sqrt{3}}{2}$$

$$\therefore Q\left(0, \frac{\sqrt{3}}{2} - \frac{\pi}{6}\right)$$

$$\therefore a \in \left(\frac{3\sqrt{3} - \pi}{6}, \infty\right) \quad \text{Ans.}$$



Q.6 (D)

$$f(x) = 5, \forall x \in [1, 2, 6]$$

$$f(1) = f(26)$$

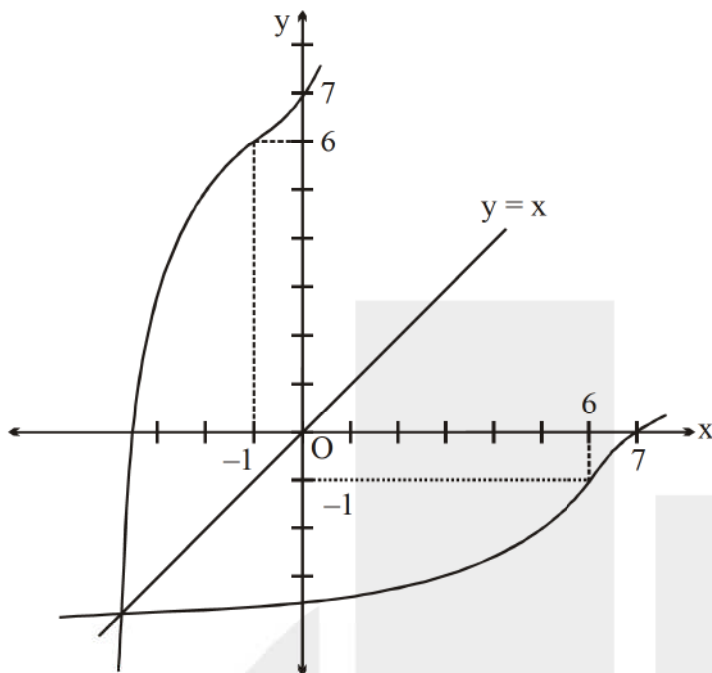
and function satisfy condition of all condition Rolle's theorem

Hence, there is atleast one real roots i.e. $f'(x) = 0$.

Q.7 (D) 1

$$x^3 + 3x^2 + 3x + 1 = (x - 6)^{1/3} - 7$$

$$\underbrace{((x+1)^3 + 6)}_{f(x)} = \underbrace{(x-6)^{1/3} - 1}_{f^{-1}(x)}$$



Clearly, $f(x)$ & $f^{-1}(x)$ has exactly one solution.

Q.8 (C)

Let $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$

Now, $f'(x) = 0 \Rightarrow x^2 - 3a \cdot x + 2a^2 = 0$

$\begin{matrix} p & q \end{matrix}$

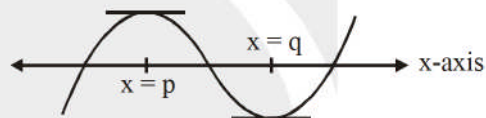
$\Rightarrow p + q = p + p^2 = 3a$

$\Rightarrow p^3 + p^6 + 3p^3(p + p^2) = 27a^3$ (cubing on both sides)

Also, $pq = p^3 = 2a^2$

$\Rightarrow 4a^2 - 9a + 2 = 0$ (As $a > 0$)

$\Rightarrow a = \frac{1}{4}, 2 \Rightarrow p = 2$ Ans. $\left(p = \frac{1}{2} \text{ (rejected)} \right)$



Q.9 (C)

$f(x) = -2x^3 - 9x^2 - 12x + 6$

$f'(x) = -6x^2 - 18x - 12$

$= -6(x^2 + 3x + 2)$

$= -6(x+1)(x+2)$

$\therefore$ Maximum value of $g(x)$ is 6. Ans.

Q.10 (C)

Let $P(x_1, y_1)$ be their point of intersection.

Also, $y_1^3 = 16x_1$

Now, $m_1 \times m_2 = -1 \Rightarrow \left(\frac{64x_1}{\alpha} \right) = 3y_1^3 \Rightarrow \alpha = \frac{4}{3}$

Q.11 (D)

Q.12 (C)

Q.13 (B)

Equation of chord: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$

Put $x = 0$

$$\therefore y = y_1 - \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x_1$$

$$\text{Hence, } y_1 - \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x_1 = 2x_1x_2 \quad [\text{Given}]$$

$$y_1x_2 - y_1x_1 - y_2x_1 + y_1x_1 = 2x_1x_2(x_2 - x_1)$$

$$\frac{y_1}{x_1} - \frac{y_2}{x_2} = 2x_2 - 2x_1$$

$$\therefore 2x_1 + \frac{y_1}{x_1} = 2x_2 + \frac{y_2}{x_2}$$

$$\text{Hence, } 2x + \frac{y}{x} = \text{constant} = C$$

curve passes through $(1, -1)$

$$2 - 1 = C \Rightarrow C = 1$$

$$2x^2 + y = x$$

$$y = x - 2x^2 \Rightarrow f(x) = x - 2x^2$$

$$(i) \quad \text{Now, } \int_0^{\frac{1}{2}} (x - 2x^2) dx = \left[\frac{x^2}{2} - \frac{2x^3}{3} \right]_0^{\frac{1}{2}} = \frac{1}{8} - \frac{1}{12} = \frac{1}{24}$$

$$(ii) \quad \text{Also, } f'(x) = 1 - 4x \geq 0 \Rightarrow x \leq \frac{1}{4}$$

$$(iii) \quad F(x) = f(x) + x = 2x - 2x^2$$

$\therefore$ Rolle's Theorem is applicable in $[0, 1]$.

Q.14 (BC)

$$g(x) = f(x) + f(1 - x)$$

$$g'(x) = f'(x) - f'(1 - x)$$

$$\therefore f''(x) < 0 \Rightarrow f'(x) \text{ is a decreasing function.}$$

$$\text{if } x > 1 - x \text{ i.e., } x > \frac{1}{2} \text{ then } f'(x) < f'(1 - x) \text{ Hence } g'(x) < 0$$

$$\Rightarrow \text{in } \left(\frac{1}{2}, 1 \right) g(x) \text{ is a decreasing function}$$

$$\Rightarrow \text{in } \left(0, \frac{1}{2} \right) g(x) \text{ is an increasing function}$$



Q.15 (BD)

$$f(x) = \frac{x(e^x - e^{-x}) + 2}{2x} = \frac{e^x - e^{-x}}{2} + \frac{1}{x}$$

$$f'(x) = \frac{e^x + e^{-x}}{2} - \frac{1}{x^2}$$

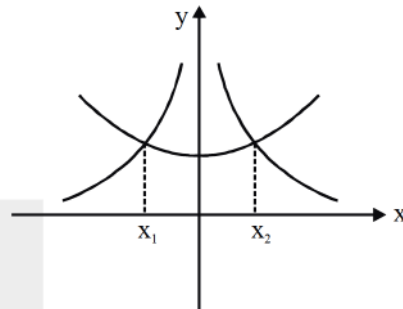
$$f'(x) = 0 \Rightarrow \frac{e^x + e^{-x}}{2} = \frac{1}{x^2}$$

$$\Rightarrow x = x_1, x_2$$

$$f''(x) = \frac{e^x - e^{-x}}{2} + \frac{2}{x^3}$$

$$\Rightarrow f''(x) > 0 \Rightarrow x \in (0, \infty)$$

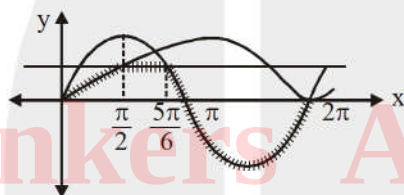
$$\text{and } f''(x) < 0 \Rightarrow x \in (-\infty, 0)$$



Q.16 (ABD)

$f(x)$ is periodic with period 2π

$$\int_0^{2\pi} f(x) dx = \left[\int_0^{\pi/2} (1 - \cos x) dx + \frac{\pi}{3} + \int_{5\pi/6}^{2\pi} 2 \sin x dx \right]$$



$$= \left[\frac{\pi}{2} - 1 + \frac{\pi}{3} - 2 \cos x \right]_0^{2\pi} = \left[\frac{5\pi}{6} - 1 - 2 \left(1 + \frac{\sqrt{3}}{2} \right) \right] = \left(\frac{5\pi}{6} - 3 - \sqrt{3} \right)$$

Clearly, $x = \pi$ is a point of inflexion of f .

Hence, number of points of inflexion in $[0, 2\pi]$ is 1.

Also, since $g'(x) = f(x)$

$\Rightarrow g$ is concave up in $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$ since slope of tangent to $y = f(x)$ intervals is positive.

Q.17 (A) P, Q, R; (B) Q, R, S; (C) P, Q, R

(A) $y = kx^2, y = e^x$ touch each other if $k = \frac{e^2}{4}$.

$\therefore$ For atleast two points in common $k > \frac{e^2}{4}$. **Ans.**

(B) $(p, q) = (\alpha, \alpha^2 + 1)$
 $(r, -4) = (-2, -4)$

$$\text{Also, } \frac{dy}{dx} = 2\alpha = \frac{-8}{-4} = 2$$

$$\alpha = 1$$

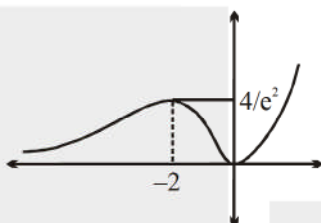
$$\therefore k \in \{1, 2, -2\}$$

$$e^k = e, e^2, \frac{1}{e^2}$$

$$(C) \quad x^2 e^x = \frac{1}{\lambda}$$

$$\frac{1}{\lambda} \in \left(0, \frac{4}{e^2}\right]$$

$$\Rightarrow \lambda \in \left[\frac{e^2}{4}, \infty\right)$$



Q.18 [0006]

Here, we must have $f'(x_0) = 0$ and $f''(x_0) = 0$ for the same $x = x_0$ (at inflection point).

$$\text{Now } f'(x) = 4x^3 + 2bx + 8$$

$$\therefore f'(x_0) = [2x_0^3 + bx_0 + 4] = 0 \quad \dots\dots(i)$$

$$\text{and } f''(x_0) = [6x_0^2 + b] = 0 \quad \dots\dots(ii)$$

From (ii), $b = -6x_0^2$, putting (i) we get

$$2x_0^3 + (-6x_0^3) + 4 = 0 \Rightarrow 4x_0^3 = 4$$

$$\Rightarrow x_0 = 1 \Rightarrow b = -6$$

Hence, $|b| = 6$ **Ans.**

Q.19 [7]

$$f(x) = \frac{x^2}{4} (2 \ln x - 1) - cx + k$$

$$f'(x) = \frac{x^2}{4} \left(\frac{2}{x}\right) + (2 \ln x - 1) \cdot \frac{x}{2} - c$$

$$= \frac{x}{2} + x \ln x - \frac{x}{2} - c$$

$$= x \ln x - c$$

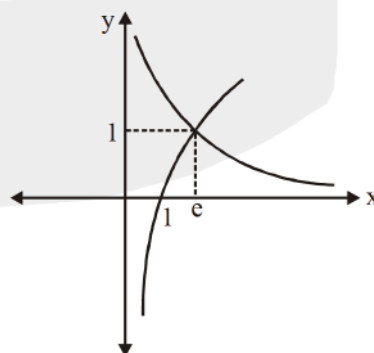
$$f'(x) = 0 \Rightarrow x \ln x - c = 0 \Rightarrow \ln x - \frac{c}{x} = 0$$

$$f'(x) < 0 \text{ in } (0, e) \Rightarrow f(x) \text{ is } \downarrow$$

$$f'(x) > 0 \text{ in } (e, \infty) \Rightarrow f(x) \text{ is } \uparrow$$

$$f(x)|_{\min.} = f(e) = \frac{e^2}{4} - e^2 + k = k - \frac{3}{4}e^2$$

For $\sqrt{f(x)}$ to be defined for all $x \in (0, \infty)$



$$f(x) \geq 0 \Rightarrow k \geq \frac{3}{4} e^2$$

$$k|_{\min.} = \frac{3e^2}{4} = \frac{ae^2}{b}$$

$$\therefore a + b = 3 + 4 = 7 \text{ Ans.}$$

Q.20 [4]

$$f(x) = 1 + 3x^2 + 5x^4 + 7x^6 + \dots + 21x^{20}, x \in \mathbb{R}$$

$$g(x) = -x^2 + 4 \cos^2 \theta - 4 \sin \theta - 7, \theta \in \mathbb{R}$$

$$f(x)|_{\min.} = 1, g(x)|_{\max.} = 4 \cos^2 \theta - 4 \sin \theta - 7 \text{ at } x = 0$$

shortest distance between $f(x)$ and $g(x)$ is

$$d = 1 - (4 \cos^2 \theta - 4 \sin \theta - 7)$$

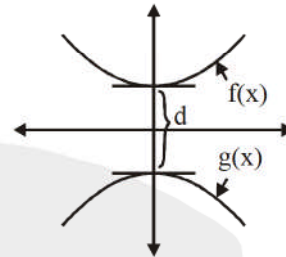
$$= 1 - (4 - 4 \sin^2 \theta - 4 \sin \theta - 7)$$

$$= 4 \sin^2 \theta + 4 \sin \theta + 4$$

$$= (2 \sin \theta + 1)^2 + 3$$

$$d_1 = 3, d_2 = 12$$

$$\therefore \frac{d_2}{d_1} = 4 \text{ Ans.}$$



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